

①

$$\mathbb{R}^n \quad \vec{x} = (x_1, x_2, \dots, x_n)$$

$$\vec{\nabla} f(\vec{x}) = \left(\frac{\partial f}{\partial x_1}(\vec{x}), \frac{\partial f}{\partial x_2}(\vec{x}), \dots, \frac{\partial f}{\partial x_n}(\vec{x}) \right)$$

Hessian $n \times n$ matrix $H_f(\vec{x})$

$$ij \text{ entry is } \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

Transpose of $\vec{y} = (y_1, y_2, \dots, y_n)$

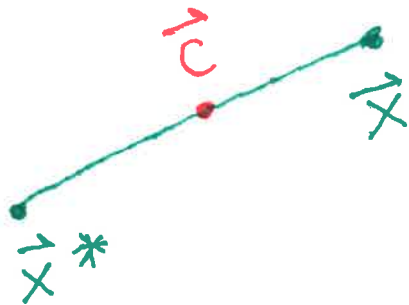
$$\vec{y}^T = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

②

$$\begin{aligned} f(\vec{x}) &= f(\vec{x}^*) + \vec{\nabla} f(\vec{x}^*) \cdot (\vec{x} - \vec{x}^*) \\ &\quad + \frac{1}{2} (\vec{x} - \vec{x}^*) H_f(\vec{x}^*) (\vec{x} - \vec{x}^*)^T \\ &\quad + \text{"cubic error" in } (\vec{x} - \vec{x}^*) \end{aligned}$$

③

$$f(\vec{x}) = f(\vec{x}^*) + \vec{\nabla} f(\vec{x}^*) \cdot (\vec{x} - \vec{x}^*) + \frac{1}{2} (\vec{x} - \vec{x}^*)^T H_f(\vec{c}) (\vec{x} - \vec{x}^*)$$



$\vec{c}$ is on segment joining $\vec{x}^*$, $\vec{x}$

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Two Dimensions

$$f(x, y) = f(x^*, y^*) + f_x(x^*, y^*)(x - x^*) \\ + f_y(x^*, y^*)(y - y^*)$$

$$+ \frac{1}{2} [f_{xx}(x^*, y^*)(x - x^*)^2 + 2f_{xy}(x^*, y^*)(x - x^*)(y - y^*) \\ + f_{yy}(x^*, y^*)(y - y^*)^2]$$

$$+ \frac{1}{3!} [f_{xxx}(c_1, c_2)(x - x^*)^3 + 3f_{xxxy}(c_1, c_2)(x - x^*)^2(y - y^*) \\ + 3f_{xyy}(c_1, c_2)(x - x^*)(y - y^*)^2 \\ + f_{yyy}(c_1, c_2)(y - y^*)^3]$$

⑤

Assume f is "smooth"

At a min (or a max)

must have $\vec{\nabla} f(\vec{x}^*) = \vec{0}$

Suppose $\vec{\nabla} f(\vec{x}^*) = \vec{0}$

$$f(\vec{x}) = f(\vec{x}^*) + \vec{\nabla} f(\vec{x}^*) \cdot (\vec{x} - \vec{x}^*)$$

$$+ \frac{1}{2} (\vec{x} - \vec{x}^*) H_f(\vec{c}) (\vec{x} - \vec{x}^*)^T$$

$$\text{If } \vec{z}^T H_f(\vec{c}) \vec{z} \geq 0 \text{ all } \vec{z}, \vec{c}$$

Then global (or abs) min at $\vec{x}^*$

$$\vec{z}^T H_f(c) \vec{z} \geq 0 \quad \text{all } z \in \mathbb{R}^n$$

⑥

$H_f(\vec{c})$ is "positive semidefinite"

[A real symmetric matrix A is positive semidefinite $\Leftrightarrow$ all eigenvalues ≥ 0] $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$

If $a > 0$ and $ac - b^2 > 0$ then A is pos. def (which implies pos semidef.)

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Lagrange Multipliers

Max or Min $f(\vec{x})$ subject to
constraint $g(\vec{x}) = c \leftarrow$ given const.

At a max or a min either (i) or
(ii) below holds

$$(i) \quad \vec{\nabla} g(\vec{x}^*) = \vec{0}$$

$$(ii) \quad \exists \lambda \in \mathbb{R} \text{ s.t. } \vec{\nabla} f(\vec{x}^*) = \lambda \vec{\nabla} g(\vec{x}^*)$$

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Ex: $f(x, y, z) = x^2 + y^2 + z^2$

Max, Min f subj to

$$g(x, y, z) = x + 2y + 3z = 28$$

$$g_x = 1, \quad g_y = 2, \quad g_z = 3 \quad \vec{\nabla} g(\vec{x}) = (1, 2, 3) \neq \vec{0}$$

$$\vec{\nabla} f(\vec{x}) = (2x, 2y, 2z) \quad \vec{\nabla} f(\vec{x}) = \lambda \vec{\nabla} g(\vec{x})$$

$$2x = \lambda$$

$$2y = 2\lambda$$

$$2z = 3\lambda$$

$$x = \lambda/2, \quad y = \lambda, \quad z = 3\lambda/2$$

$$\frac{\lambda}{2} + 2\lambda + 3\left(\frac{3\lambda}{2}\right) = 28$$

$$\lambda + 4\lambda + 9\lambda = 56 \quad \lambda = 4$$

⑨

$$x^* = 2, \quad y^* = 4, \quad z^* = 6$$

Global Min

(No Max)

Ex 2: Max, Min $f(x, y) = xy$

subj to $g(x, y) = \frac{1}{2}x^2 + 2y^2 = 9$

$$f_x = y, \quad f_y = x, \quad g_x = x, \quad g_y = 4y$$

$$\vec{\nabla} g(x, y) = \vec{0} \iff x = y = 0 \text{ impossible}$$

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$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

$$\begin{cases} y = \lambda x \\ x = 4\lambda y \end{cases}$$

$$x = 4\lambda(\lambda x) = 4\lambda^2 x$$

Can't have $x=0$

(because then $y=0$
and constraint is
violated)

$$4\lambda^2 = 1, \quad \lambda = \pm \frac{1}{2}$$

$$\lambda = \frac{1}{2} \quad y = \frac{1}{2}x \quad \frac{1}{2}x^2 + 2\left(\frac{1}{2}x\right)^2 = 9$$

$$x^2 = 9, \quad x = \pm 3$$

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$(3, \frac{3}{2}), (-3, -\frac{3}{2})$ Both
Maximizers

$$\lambda = -\frac{1}{2}, \quad y = -\frac{1}{2}x$$

$$x = \pm 3 \quad (3, -\frac{3}{2}), (-3, \frac{3}{2})$$

Both Minimizers

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Chain Rule

1st Case: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ has cont. 1st partial derivs, $I \subset \mathbb{R}$ is an interval, $\vec{u}: I \rightarrow \mathbb{R}^n$ is diff. $[F: I \rightarrow \mathbb{R}$
 $F(t) = f(\vec{u}(t))]$

Put $F(t) = f(\vec{u}(t))$. Then F is diff
and

$$F'(t) = \vec{\nabla} f(\vec{u}(t)) \cdot \vec{u}'(t)$$

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$$W = f(x_1, x_2, x_3, \dots, x_n)$$

$$x_1 = u_1(t), x_2 = u_2(t), \dots, x_n = u_n(t)$$

$$\frac{dW}{dt} = \frac{\partial f}{\partial x_1} u_1'(t) + \frac{\partial f}{\partial x_2} u_2'(t) + \dots + \frac{\partial f}{\partial x_n} u_n'(t)$$

$$= \frac{\partial W}{\partial x_1} \frac{dx_1}{dt} + \frac{\partial W}{\partial x_2} \frac{dx_2}{dt} + \dots + \frac{\partial W}{\partial x_n} \frac{dx_n}{dt}$$

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$$\vec{\mu}(s, t)$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$F(s, t) = f(\vec{\mu}(s, t))$$

$$\frac{\partial F}{\partial s} = \vec{\nabla} f(\vec{\mu}(s, t)) \cdot \frac{\partial \vec{\mu}}{\partial s}$$

$$\frac{\partial F}{\partial t} = \vec{\nabla} f(\vec{\mu}(s, t)) \cdot \frac{\partial \vec{\mu}}{\partial t}$$

$$W = f(x_1, x_2, \dots, x_n)$$

$$x_i = u_i(s, t), \quad i = 1, 2, \dots, n$$

$$\frac{\partial W}{\partial s} = \frac{\partial W}{\partial x_1} \frac{\partial x_1}{\partial s} + \frac{\partial W}{\partial x_2} \frac{\partial x_2}{\partial s} + \dots + \frac{\partial W}{\partial x_n} \frac{\partial x_n}{\partial s}$$

$$\frac{\partial W}{\partial t} = \frac{\partial W}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial W}{\partial x_2} \frac{\partial x_2}{\partial t} + \dots + \frac{\partial W}{\partial x_n} \frac{\partial x_n}{\partial t}$$

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$$w = f(x_1, x_2, \dots, x_n)$$

$$\frac{\partial w}{\partial \square} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial \square} + \frac{\partial w}{\partial x_2} \frac{\partial x_2}{\partial \square} + \dots + \frac{\partial w}{\partial x_n} \frac{\partial x_n}{\partial \square}$$

Put any variable in $\square$

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$$\text{Ex: } Z = X_2^3 e^{5X_1} + X_3^4$$

$$X_1 = \sin \alpha + 2\beta^2, \quad X_2 = \alpha^2 + \beta^2 + \gamma^2, \quad X_3 = \beta^{10}$$

$$\begin{aligned} \frac{\partial Z}{\partial \alpha} &= \frac{\partial Z}{\partial X_1} \frac{\partial X_1}{\partial \alpha} + \frac{\partial Z}{\partial X_2} \frac{\partial X_2}{\partial \alpha} + \frac{\partial Z}{\partial X_3} \frac{\partial X_3}{\partial \alpha} \quad \begin{matrix} \nearrow 0 \\ \nearrow 0 \end{matrix} \\ &= 5X_2^3 e^{5X_1} (\cos \alpha) + 3X_2^2 e^{5X_1} (2\alpha) + 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial Z}{\partial \gamma} &= \frac{\partial Z}{\partial X_1} \frac{\partial X_1}{\partial \gamma} + \frac{\partial Z}{\partial X_2} \frac{\partial X_2}{\partial \gamma} + \frac{\partial Z}{\partial X_3} \frac{\partial X_3}{\partial \gamma} \quad \begin{matrix} \nearrow 0 \\ \nearrow 0 \end{matrix} \\ &= 3X_2^2 e^{5X_1} (2\gamma) \end{aligned}$$

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Total Time Derivative

$f(t, x, y)$ x, y fns of t $\frac{d}{dt}(t)$

$$\begin{aligned}\frac{d}{dt} f(t, x(t), y(t)) &= f_{,1}(t, x(t), y(t)) \cdot 1 \\ &\quad + f_{,2}(t, x(t), y(t)) \cdot x'(t) \\ &\quad + f_{,3}(t, x(t), y(t)) \cdot y'(t)\end{aligned}$$

Sometimes Written

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

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Ex: $f(t, x) = e^{2t} \sqrt{t+x^2}$

$$X(0) = 1, \quad X'(0) = -3$$

Compute $\left. \frac{d}{dt} f(t, X(t)) \right|_{t=0}$

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$$f_{,1}(t,x) = 2e^{2t} \sqrt{t+x^2} + e^{2t} \left(\frac{1}{2\sqrt{t+x^2}} \right)$$

$$f_{,2}(t,x) = \frac{e^{2t} x}{\sqrt{t+x^2}}$$

$$\left. \frac{d}{dt} f(t, x(t)) \right|_{t=0} = f_{,1}(0,1) + f_{,2}(0,1)(-3)$$

$$= 2 + \frac{1}{2} - 3 = -\frac{1}{2}$$

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Another Lagrange Example

Max, Min $f(x,y) = 2x^2 + y^2$

Subj to $3x - 2y = 6$

$$f_x = 4x$$

$$f_y = 2y$$

$$g_x = 3$$

$$g_y = -2$$

$$\vec{\nabla} g \neq (0,0)$$

So look for
 λ

$$\vec{\nabla} f = \lambda \vec{\nabla} g$$

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$$4x = 3\lambda, \quad 2y = -2\lambda$$

$$x = \frac{3}{4}\lambda, \quad y = -\lambda$$

$$3x - 2y = 6$$

$$\frac{9}{4}\lambda + 2\lambda = 6$$

$$\frac{17}{4}\lambda = 6$$

$$\lambda = \frac{24}{17}$$

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$$x^* = \frac{3}{4} \left(\frac{24}{17} \right)$$

$$y^* = -\frac{24}{17}$$

$$= \frac{18}{17}$$

$$H_f(x) = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

Pos Semi
Def.

$$(z_1, z_2) \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = 4z_1^2 + 2z_2^2$$

$$\begin{pmatrix} 4z_1 \\ 2z_2 \end{pmatrix}$$

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Constraints Linear

Absolute Min